

Machine Learning in Physics

Big Ideas from 6 Representative Papers — Survey & Course Capstone

PHYG004 — 2026 Spring · Lecture 20 (Final Lecture)

Prof. Young Woo Choi · Department of Physics, Sogang University

No heavy math — just the core idea of each paper, one figure and one equation at a time. Plus: how everything from L01–L19 connects here.

6/5/2026

What You Have Learned This Semester

A 20-lecture journey through ML for physics — three blocks, one through-line.

Foundations (L01–L10)

JAX · autodiff · NN · CNN ·
PINN · differentiable physics
· transformers

Key spine: QHO, Ising, double-well
introduced

Symmetry & Structure (L11–L13)

GNN · e3nn · equivariance ·
ML potentials

Geometric DL blueprint:
CNN → GNN → e3nn

Generative & Inference (L14–L19)

VAE · NF · Flow Matching ·
Diffusion · EBM · SBI

Generative arc from ELBO to
posterior estimation

Today: six landmark papers show where these ideas land in research practice.

The Big Picture: AI Is Learning to Do Physics

Across very different papers, neural networks are being taught to do **three jobs**:



1. SAMPLE

Generate physical states that follow nature's odds

Boltzmann Generators · VAN
+ EBM Langevin (L18) · SBI posterior (L19)



2. READ

Understand & compress the quantum electronic structure

Kohn–Sham VAE · DeepH · MBFormer



3. INVENT

Design brand-new materials to order

MatterGen · EDM
diffusion (L17) + equivariant (L11–12)

One recurring trick: a neural network is just a **very flexible function** we fit — here we fit it to **physics**.

The Generative Family — Where Each Method Lives

This semester's generative arc maps onto the Sample/Read/Invent jobs:

Continuous structures — sampling trajectories

Method	Lecture	Spine system
Normalizing Flow (RealNVP)	L15	double-well / alanine
Flow Matching	L16	alanine dipeptide
Diffusion (DDPM/score)	L17	alanine dipeptide

Latent structure — reading & generating

Method	Lecture	Physics job
VAE / ELBO	L14	compress wavefunctions
Autoregressive	L20 paper 2	spin configurations
Equivariant diffusion	L20 paper 6	crystal generation

Job 1 — Sampling

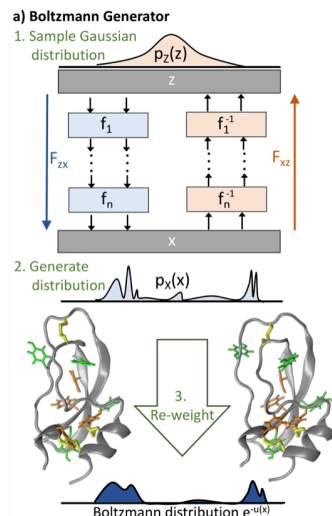
Teaching AI to roll the dice the way nature does

Boltzmann Generators · Variational Autoregressive Networks

1 · Boltzmann Generators — the idea

Noé, Olsson, Köhler, Wu · Science 365, eaaw1147 (2019) · Normalizing flow

A molecule jiggles between shapes; likely shapes have low energy (the **Boltzmann law**). Normal simulation crawls one tiny step at a time — painfully slow.



💡 An AI prints equilibrium molecule snapshots in one shot — then fixes any bias exactly.

🎯 Like a slot machine that instantly prints molecule poses, each with a "fairness ticket" saying how much it really counts.

Reading the figure:

- Top: draw a random point from a simple bell curve
- Arrows: a **reversible** network turns it into a real molecule shape
- Bottom: **re-weight** so the shapes obey $e^{-u(x)}$

1 · Boltzmann Generators — the equation

$$w(x) = \frac{e^{-u(x)}}{p_X(x)}$$

Read it as: **weight** = (how likely physics says this shape is) ÷ (how likely the AI made it)

What each symbol means

$w(x)$	how much this shape counts
x	a molecule shape
$u(x)$	its energy
$e^{-u(x)}$	Boltzmann likelihood (target)
$p_X(x)$	chance the AI made it

Why it matters

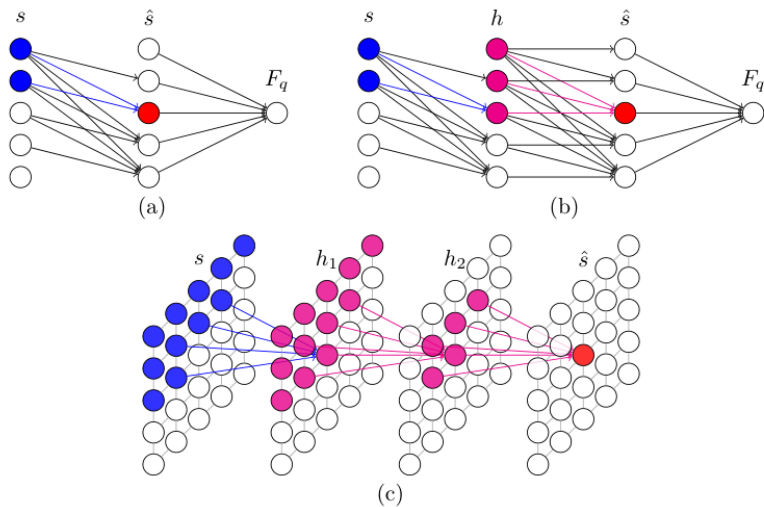
If the AI made a shape too often, p_X is big, so w shrinks it back down — and vice-versa. The weights make the answer **exactly unbiased**.

→ Free energies of proteins & materials **without** waiting through slow rare events.

2 · Variational Autoregressive Networks — the idea

Wu, Wang & Zhang · Phys. Rev. Lett. 122, 080602 (2019) · Lei Wang group · Autoregressive net

For many interacting spins (tiny ± 1 magnets) we want the **free energy** F . Listing all 2^N arrangements is hopeless.



💡 Build a network whose training loss *is* the free energy — so learning = minimizing F .

🎯 Like guessing a deck's order one card at a time, each guess using the cards already shown.

Reading the figure:

- Inputs = spins s ; output = the network's guess $\hat{s}$
- Each output connects **only to earlier** spins
- from one layer (a) up to a 2D-lattice net (panel c)

2 · Variational Autoregressive Networks — the equation

$$F_q = \langle E(s) \rangle_{q_\theta} - T S[q_\theta]$$

Read it as: **the network's score = average energy – temperature × entropy = the free energy itself**

What each symbol means

F_q	free energy estimate = the loss
s	a configuration (all spin values)
$E(s)$	its energy
q_θ	network's probability (weights θ)
$\langle E \rangle_{q_\theta}$	average energy over its samples
$S[q_\theta]$	entropy (spread) of the network

Why it matters

This is the bridge of the whole course: **a machine-learning loss and a physics free energy are the same equation.**

Same idea as VAE ELBO (L14): minimizing the loss = minimizing a variational free energy.

→ Turns a hard statistical-mechanics problem into network training — exact bound on F , fast

Job 2 — Reading

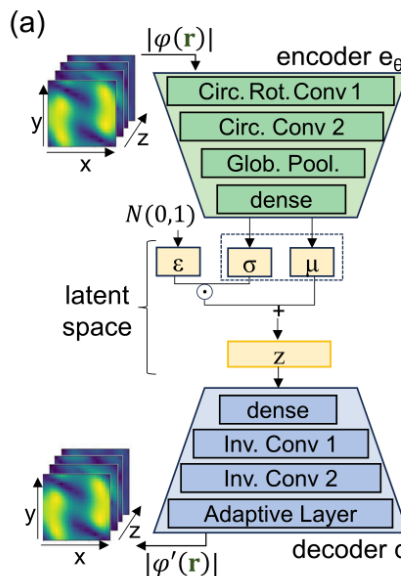
Teaching AI to understand the quantum electronic
structure

Kohn-Sham VAE · DeepH · MBFormer

3 · A VAE for Wavefunctions — the idea

Hou, Wu & Qiu · Nat. Commun. 15, 9481 (2024) · Variational Autoencoder (VAE)

A quantum wavefunction from **DFT** is stored as a huge grid of numbers. Can we boil it down to a few essential numbers without losing the physics?



💡 A wavefunction looks huge but is really just a handful of numbers in disguise.

🎧 Like an MP3: raw audio is millions of samples, but the file keeps only the few numbers that matter.

Reading the figure:

- Top: wavefunction enters the green **encoder**
- Middle: squeezed to a tiny code z (bottleneck)
- Bottom: blue **decoder** rebuilds it — nearly identical

Architecture connection: VAE / ELBO (L14). The latent code z is the order-parameter-like collective coordinate.

3 · A VAE for Wavefunctions — the equation

$$\mathcal{L} = \underbrace{\|\phi - d_{\theta}(e_{\theta}(\phi))\|^2}_{\text{rebuild error}} + \beta \underbrace{D_{KL}}_{\text{tidy code}}$$

Read it as: **cost = (how badly we rebuilt the wavefunction) + (a tidiness penalty on the code)**

What each symbol means

$\mathcal{L}$	total training cost to minimize
ϕ	input wavefunction (big grid)
e_{θ}, d_{θ}	encoder (squeeze) & decoder (rebuild)
$\ \cdot\ ^2$	mismatch between input & rebuilt copy
D_{KL}	keeps the code smooth & normal
β	knob: accuracy vs. tidy code

Why it matters

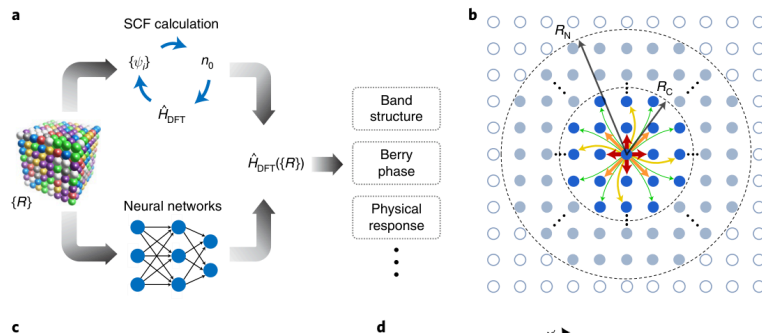
The wavefunction collapses to **~30 numbers** with little loss — proof it secretly lives on a *low-dimensional surface*.

→ Those few numbers predict expensive **many-body (GW) bandstructures** to ~0.11 eV.

4 · DeepH: Learning the Hamiltonian — the idea

He Li et al. · Nat. Comput. Sci. 2, 367 (2022) · Equivariant graph neural network

DFT finds a material's electrons by slowly solving Schrödinger's equation over and over — accurate, but too costly for many thousands of atoms.



💡 Teach an AI to *write down* the quantum Hamiltonian matrix straight from atom positions — skip the slow solver.

🎯 Instead of re-solving a hard puzzle each time, a student learns how the answer depends on the setup and just fills it in.

Reading the figure: Left (a) — the slow **self-consistent DFT loop**; the lower path replaces it with a neural network. Right (b) — **locality**: only atoms inside the cutoff R_C matter, so the job stays small.

4 · DeepH: Learning the Hamiltonian — the equation

$$\{\mathcal{R}\} \xrightarrow{\text{neural network}} \hat{H}_{\text{DFT}}(\{\mathcal{R}\})$$

Read it as: **give the network where the atoms are; it returns the quantum Hamiltonian matrix**

What each symbol means

$\{\mathcal{R}\}$	the set of all atom positions
$\mapsto$	"maps to" — what the network learns
$\hat{H}_{\text{DFT}}$	the DFT Hamiltonian (energy operator)

The Hamiltonian depends **only on where the atoms sit**, so locality makes it learnable piece-by-piece.

How it works

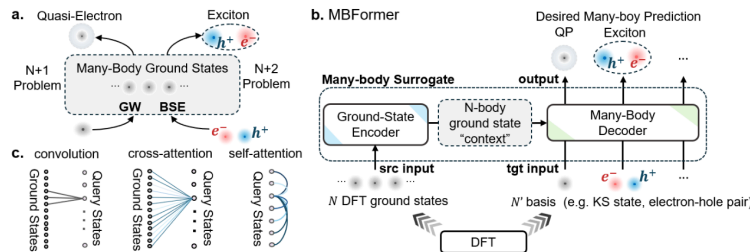
- Atoms = graph nodes, near-neighbors = edges
- Pass messages between nearby atoms
- Output the Hamiltonian block for each pair
- Local coordinates $\rightarrow$ rotates correctly with the structure

Why it matters $\rightarrow$ A fast lookup replaces costly DFT, making giant systems (e.g. magic-angle

5 · MBFormer: Attention for Electrons — the idea

Hou, Xu, Wu & Qiu · arXiv:2507.05480 (2025) · Transformer

Cheap **DFT** gives single-electron states but misses how electrons **interact** (the "many-body" problem). Accurate methods (GW-BSE) for band gaps & light absorption are very slow.



💡 The same "attention" trick behind ChatGPT lets electron states *talk to each other* to predict how they interact.

🎯 To predict someone's behavior in a room, watch how everyone attends to everyone else — not each person alone.

Reading the figure: N cheap DFT states enter the **Ground-State Encoder**, squeezed into a shared "context." A chosen **target** (an electron, or an electron-hole pair) enters the **Decoder**, which looks back at that context to predict its many-body energy. Bottom: *attention* links a state to **all** others at once.

5 · MBFormer: Attention for Electrons — the equation

$$H_{ij}^{MB} = \langle i | \hat{H}^{MB}(\{\phi_{n\mathbf{k}}, \epsilon_{n\mathbf{k}}\}) | j \rangle$$

Read it as: **each many-body table entry is built from the cheap DFT states and their energies**

What each symbol means

H_{ij}^{MB}	one entry of the many-body table
i, j	two chosen electron/hole states
$\hat{H}^{MB}$	the many-body operator to learn
$\phi_{n\mathbf{k}}$	a DFT single-electron state
$\epsilon_{n\mathbf{k}}$	that state's DFT energy
$n, \mathbf{k}$	band index & crystal momentum

Why it matters

Slow excited-state physics — **band gaps, optical absorption, excitons** — becomes a fast prediction.

→ Hints at a single **foundation model** for materials physics, the way one model now handles language.

Job 3 — Inventing

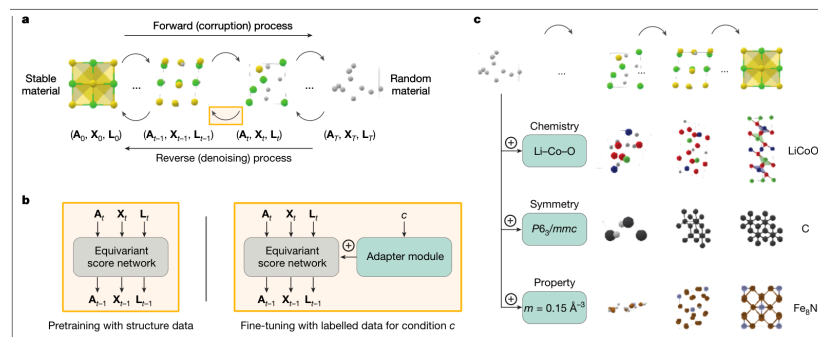
Teaching AI to design brand-new materials

MatterGen · EDM — diffusion meets equivariance

6 · MatterGen: Inventing Crystals — the idea

Zeni et al. (Microsoft) · Nature 639, 624 (2025) · Diffusion model [published PDF via Sogang proxy]

Finding new **stable** crystals (atoms in their lowest-energy arrangement) is slow. We want a computer to propose useful new ones — with chosen properties like magnetism.



💡 **AI invents crystals by *cleaning up* random atoms into a stable structure, step by step.**

🎯 *Like un-shuffling a deck: scramble a real crystal into noise, then teach the computer to reverse every shuffle.*

Reading the figure: Top arrow = **corrupt** a real crystal into a random cloud. Bottom arrow = the model **denoises** it back, fixing atom types A , positions X , and lattice L together. Right: small **adapters** steer it toward a target property (e.g. LiCoO_2 , Fe_3N).

6 · MatterGen: Inventing Crystals — the equation

$$(A_0, X_0, L_0) \xrightarrow{\text{corrupt}} (A_T, X_T, L_T) \xrightarrow{\text{denoise}} (A_0, X_0, L_0)$$

Read it as: **scramble a crystal into noise, then learn to run the scramble backwards**

What each symbol means

A	atom types (which elements)
X	atom positions in the cell
L	lattice (shape & size of the box)
(A_0, X_0, L_0)	a real, stable crystal
(A_T, X_T, L_T)	pure noise (random atoms)
T	number of small steps

Why it matters

Start from noise, denoise → a **fresh, never-seen, stable crystal**. Adapters aim it at target chemistry or properties.

→ One generated material, **TaCr₂O₆**, was actually synthesized — property within 20% of the target.

6b · EDM: Equivariant Diffusion for Molecules

Hoogeboom, Satorras, Vignac & Welling · ICML 2022 · Equivariant diffusion

MatterGen generates crystals (periodic). **EDM** generates *molecules* (finite). The key challenge: a molecule's energy is the same no matter how you rotate or translate it — the generative model must respect this.

💡 **Run a diffusion model entirely in equivariant coordinates — so the generated molecule has the right physics symmetry built in.**

🌀 *Don't just learn to un-shuffle; learn to un-shuffle in a way that never cares which direction "up" is.*

Key idea: the noise process and denoising network both operate in **E(3)-equivariant** space (positions transform as vectors; atom features as scalars or higher-order irreps).

Why equivariance matters here

If we diffuse in raw Cartesian $\mathbb{R}^3$, the model wastes capacity learning that a rotated version of the same molecule is equally valid.

EDM enforces this: the score network $s_\theta(x_t, t)$ satisfies

$$s_\theta(Rx_t, t) = R s_\theta(x_t, t)$$

for any rotation $R \in SO(3)$.

Architecture connection: E(3)-equivariant GNN (L11–L12) inside a diffusion (L17) framework — the

The Named Spine: Where Benchmark Systems Reappeared

From L01's declaration to L20's survey — two systems ran through the whole semester.

Ising Model

Lecture	Role
L01	JAX intro — Ising MC
L06	CNN phase classification
L14	Ising-VAE, latent $\leftrightarrow$ magnetization
L20 (paper 2)	VAN — autoregressive free energy

Double-well / Alanine Dipeptide

Lecture	Role
L02	JAX double-well gradient descent
L15	NF Boltzmann target (RealNVP)
L16	Flow Matching velocity field
L17	DDPM reverse process

From-Scratch vs Foundation Model: A Course-Level Takeaway

Use a public checkpoint

When: your domain matches the pretraining distribution; inference only needed

Examples:

- MACE-MP-0 zero-shot on new elements
- MatterGen generating stable crystals
- GPT-2 for physics text

Fine-tune a checkpoint

When: domain shift is moderate; labeled data is scarce (<1k); pretraining too expensive

Examples:

- MACE-MP-0 + ~200 Si DFT configs (L13)
- ResNet-18 + Galaxy10 (L07)
- GPT-2 + physics corpus (L10)

Train from scratch

When: strong physics constraints to enforce; unusual input modality; foundation model unavailable

Examples:

- DeepH (equivariant Hamiltonian)
- PINN (PDE constraints)
- VAN (Boltzmann variational loss)

Active Mapping: Connect the Papers to Your Lectures

For each paper, identify the architecture family you learned and the lecture where you first met it.

Paper	Architecture	Lecture	Core loss / sampling	Physics target
Boltzmann Generators	Normalizing Flow	L15	Importance weight $w = e^{-u}/p_X$	Alanine dipeptide
VAN	Autoregressive	—	$F_q = \langle E \rangle - TS$ = free energy	Ising spins
Kohn–Sham VAE	VAE	L14	ELBO / $-D_{KL}$ + reconstruction	DFT wavefunctions
DeepH	Equivariant GNN	L11–12	MSE on H matrix elements	Hamiltonian
MBFormer	Transformer	L10	Cross-attention regression	Many-body energies
MatterGen	Equivariant diffusion	L11–12 + L17	Score / DDPM denoising	Crystal generation
EDM	Equivariant diffusion	L11–12 + L17	SE(3)-equivariant score	Molecule generation

Takeaways — the one slide to remember

- A neural network is a **flexible function** we fit. These papers fit it to **physics**, doing three jobs:
- 🎲 **Sample** — *Boltzmann Generators* & *VAN* generate states with nature's odds. Big idea: **the ML loss can literally be the free energy**. (Also: EBM Langevin L18, SBI posterior L19.)
- 🔍 **Read** — *Kohn–Sham VAE* compresses wavefunctions, *DeepH* predicts the Hamiltonian, *MBFormer* uses attention for many-body effects. Big idea: **the quantum electronic structure is simpler & more local than it looks**.
- 🖋️ **Invent** — *MatterGen* & *EDM* denoise random atoms into new materials on demand. Big idea: **diffusion + equivariance = physically valid generation**.
- The recurring theme: **loss $\leftrightarrow$ energy**, and **let physics (symmetry, locality, the Boltzmann law) shape the network**.
- Flow Matching (L16) is the *normalizing-flow sibling* of diffusion — it learns a straight velocity field instead of a denoising score, but reaches the same Boltzmann target.

You don't need to build these to use them — but now you know **what each one is really doing**.

Term Project Kick-Off

Use the taxonomy to position your project. Answer these four questions:

1. **Which job?** Is your project about *sampling* a physical distribution, *reading* (compressing or predicting) a quantum/classical structure, or *inventing* new configurations/materials?
2. **Which architecture?** Which paper in today's survey most resembles your approach? Map it to a lecture (L14–L19) where you saw the method live.
3. **Which data level?** Toy/analytic · simulated (MD/MC) · real measured/computed (DFT, experiment)?
4. **From scratch or fine-tune?** Is there a publicly available checkpoint (MACE-MP-0, MatterGen, SevenNet) you can adapt, or does your physics constraint require training from scratch?

Write your answers on one page and bring them to the first project check-in (Week 15). The four questions become your project proposal outline.

References & Links

1. F. Noé, S. Olsson, J. Köhler, H. Wu, *Boltzmann Generators: Sampling Equilibrium States of Many-Body Systems with Deep Learning*, **Science** **365**, eaaw1147 (2019).
2. D. Wu, L. Wang, P. Zhang, *Solving Statistical Mechanics Using Variational Autoregressive Networks*, **Phys. Rev. Lett.** **122**, 080602 (2019). — *Lei Wang group*, [*scholar profile*](#)
3. B. Hou, J. Wu, D. Y. Qiu, *Unsupervised representation learning of Kohn–Sham states ...*, **Nat. Commun.** **15**, 9481 (2024).
4. He Li et al., *Deep-learning density functional theory Hamiltonian ... (DeepH)*, **Nat. Comput. Sci.** **2**, 367 (2022).
5. B. Hou, Y. Xu, J. Wu, D. Y. Qiu, *MBFormer: A Transformer-based Learning Paradigm for Many-body Interactions*, **arXiv:2507.05480** (2025).
6. C. Zeni et al., *A Generative Model for Inorganic Materials Design (MatterGen)*, **Nature** **639**, 624 (2025).
7. E. Hoogetboom, V. G. Satorras, C. Vignac, M. Welling, *Equivariant Diffusion for Molecule Generation in 3D (EDM)*, **ICML 2022**.

Figures reproduced from the cited papers for educational use. · End of course — thank you! 🎓

Next: Week 15 term-project presentations — bring your four-question project proposal.